A filter isn't access control

Day 05: a local vector index, and one managers-only chunk.

Q: When do pay rises take effect?

`manager-pay-review-guide#overview` #1 · 0.70

only managers, people-team · employee + filter: not returned

Same index, same question. Four ways it leaks:

1 FILTER FORGOTTEN

`search(q) # where=None`

a debug path, a widget, an eval:
no error, results look normal

pay guide back at #1

2 GROUPS FROM THE REQUEST

`groups = ['managers']`

an employee claims a group:
same call as a manager

pay guide back at #1

3 PERMISSIONS STALE

travel policy made managers-only,
index not synced yet
(the fix: 5 records, 0 re-embeds)

employee still gets it

4 THE FILE ITSELF

`np.load('index.npz')`

all 44 texts, no filter runs,
plus vectors that can leak text

pay guide text readable

WHAT MAKES IT HOLD

deeper: Day 14

- ✓ groups from the directory
- ✓ unknown user → nothing
- ✓ filter can't be skipped
- ✓ permissions synced fast

Vectors + records + the model

What a vector index stores, and where the filter goes.

THE INDEX: ONE FILE, index.npz (99,192 bytes)



vectors

44 × 384 float32

+



records

1 per row, key chunk_id

+



the model

bge-small @ 5c38ec7c405e

chunk_id	manager-pay-review-guide#overview
chunk_hash	146c27100b941450...
allowed_groups	managers, people-team
last_reviewed	None

FILTER BEFORE YOU RANK

Employee asks: "When do pay rises take effect?"

no filter	0.70	0.59	0.58	top 3
post-filter	×	0.59	0.58	2 of 3
pre-filter	0.59	0.58	0.57	3 of 3

"How do I connect to the VPN?" · scope: remote work policy

no filter	0.80 VPN guide	0.76 VPN guide	0.74 VPN guide	top 3
post-filter	×	×	×	0 of 3
pre-filter	0.71	0.51	0.51	3 of 3

Sync, don't just upsert

Day 02's changes + a renamed heading, applied to the saved index.

SYNC PLAN, BY chunk_id AND chunk_hash

add	2	new policy · renamed heading
re-embed	1	USD 500 → USD 600
update metadata	11	contractors added · new doc version
delete	5	retired VPN guide · old heading
unchanged	27	incl. a line-ending-only re-save

3 of 41 texts through the model · same result as a rebuild

Q: How do I connect to the VPN? (the VPN guide was retired)

UPSERT ONLY — 46 chunks

#1 0.80 vpn-access-guide #troubleshooting

#2 0.76 vpn-access-guide #connecting

#3 0.74 vpn-access-guide #when-to-use-the-vpn

retired guide fills the top 3

FULL SYNC — 41 chunks

#1 0.71 remote-work-policy #security-while-remote

#2 0.58 new-starter-it-checklist #your-first-day

#3 0.57 password-and-mfa-standard #multi-factor-authentication

remote work policy's VPN advice

ATOMIC SAVE

save fails halfway:

naive → file broken (EOFError)

atomic → old index loads (44)

DELETE GUARD

ingest returns nothing:

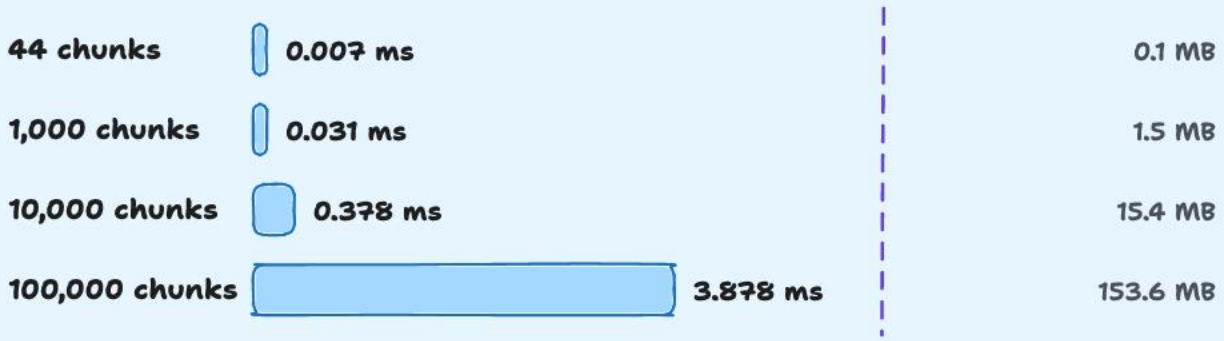
"refusing to delete 44 of 44 chunks"

Do you need a vector DB?

Exact NumPy search, measured, vs the cost of adding a library.

EXACT TOP 3, PER QUESTION

2-vCPU cloud VM



INSTALL SIZE ON TOP OF NUMPY

measured 2026-10-18



Even 100,000 chunks searched faster than the model embeds one question. No library yet.
Carries over to any vector DB: records, filters, sync.

Next → Day 06: grounded answers with citations